

FL Simulation Results

File: test_A8

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 95.0% (0.95)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 95.0% (0.95) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8736
Clean + DP Protection Accuracy: 0.8740
Poisoned Accuracy: 0.7543
Poisoned + DP Protection Accuracy: 0.7894
Drop (Clean - Poisoned): 0.1193
Drop (Clean - Init): 0.0808
Drop (Clean DP - Init): 0.0812
Drop (Poisoned - Init): -0.0385
Drop (Poisoned DP - Init): -0.0034
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8798

Clean Recall: 0.8736

Clean F1 Score: 0.8729

Clean DP Precision: 0.8798

Clean DP Recall: 0.8740

Clean DP F1 Score: 0.8734

Poisoned Precision: 0.7832

Poisoned Recall: 0.7543

Poisoned F1 Score: 0.7561

DP Protection Precision: 0.8020

DP Protection Recall: 0.7894

DP Protection F1 Score: 0.7873

Summary

FL Simulation Complete

Task: bc80ba61-d53e-45db-81f2-3e237a1b9ed4

GPU: GPU 0

Init Accuracy: 0.7928

Clean Accuracy: 0.8736

Clean DP Accuracy: 0.8740

Poisoned Accuracy: 0.7543

Data Poison Protection Accuracy: 0.7894

Drop (Clean - Poisoned): 0.1193

Drop (Clean - Init): 0.0808

Drop (Clean DP - Init): 0.0812

Drop (Poisoned - Init): -0.0385

Drop (Poisoned_DP - Init): -0.0034

Data Poison Protection Method: foolsgold

--- Confusion Matrix Metrics (Weighted Avg) ---

Clean Precision: 0.8798

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Poisoned Precision: 0.7832

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DP Protection Precision: 0.8020

DP Protection Recall: 0.7894

DP Protection F1 Score: 0.7873